

ESTIMATION PRO — FIXED RATE BAND COST ESTIMATE

Project: Quick Analysis

Generated: 2026-05-27T03:34:11.109317+00:00

Model: STRUCTMIND CORE · PRO

SHA-256: 3bcebbf2a98fc21496b41d4dfe7774a94cd4d18832bebc7d74729f1a488ba520

CALCULATION MANIFEST — SINGLE SOURCE OF TRUTH

--- All locked input and output values for this estimate. Sections 1 and 7 copy from here.

Parameter	Value
AI-Extracted Fabricated Tonnage	37.08 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Material Factor	x1.00
Surface Factor	x1.00
Assembly Factor	x1.10
Composite Adjustment Factor	x1.10
Fixed Rate — Low (USD per ton)	\$3,500
Fixed Rate — Mid (USD per ton)	\$4,000
Fixed Rate — High (USD per ton)	\$4,500
Adjusted Rate — Low	\$3,850
Adjusted Rate — Mid	\$4,400

Parameter	Value
Adjusted Rate — High	\$4,950
Country	USA
Tax Rate Applied	0%
LOCK COST — LOW	\$142,800
LOCK COST — MID	\$163,200
LOCK COST — HIGH	\$183,500

1. EXECUTIVE SUMMARY

--- Project cost summary at fixed rate band.

Metric	Value
Project Name or ID	HOME 2 - CAROWINDS
Country	USA
Total Fabricated Tonnage	37.08 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Composite Adjustment Factor	x1.10
Fixed Rate Band Applied	\$3,500 — \$4,500 per ton
Tax Rate	0%
Estimated Cost — Low (inc. tax)	\$142,800
Estimated Cost — Mid (inc. tax)	\$163,200

Metric	Value
Estimated Cost — High (inc. tax)	\$183,500
Cost per Fabricated Ton — Mid	\$4,400 per ton

Top Three Commercial Risks: 1. CRITICAL: All steel lintels (W-shapes and angles) are entirely unquantified. Schedules on S0.02 refer to Architectural drawings which were not provided. This uncoded scope could add 5-10 tons. 2. CRITICAL: Column lengths are not specified in the Steel Column Schedule (S0.02). A full building height of 43'-0" was assumed for all primary columns based on elevations. Incorrect lengths will significantly alter tonnage. 3. WARNING: Several HSS profiles (HSS18x6x3/8, HSS9x5x3/8, HSS6x6x5/16, HSS6x6x5/8, HSS8x6x3/8) are not in the provided AISC unit weight table. Weights were sourced externally and require verification.

2. BASIS OF ESTIMATE

--- Drawings reviewed: S0.01, S0.02, S1.01, S1.02, S1.03, S1.04, S2.01, S2.02, S2.03, S2.04, S3.01, S3.02, S3.03, S3.04, S3.05, S3.06.

Tonnage extraction method: (c) Estimated from framing plans where BOM is absent or incomplete. All quantities and lengths for beams and columns were counted and scaled directly from framing plans S1.01 and S1.02. Schedules on S0.02 were used for member profiles and plate sizes but were critically incomplete, lacking quantities and lengths.

Composite factor selection:

- Material (Carbon Steel): Explicitly stated in General Notes on S0.01 (ASTM A992, A500, A36).
- Surface Treatment (Shop primer only): Stated as the primary finish in General Note 11 on S0.01. The requirement for galvanized bearing plates and zinc-rich primer on some beam ends is noted as a risk but assumed to be covered within the rate band.
- Assembly Complexity (Mixed bolt and weld): Assumed from various details. S0.02 shows a bolted Wind Post connection, while the Lintel Schedule implies welded bottom plates. Canopies and trellises (S3.05, S3.06) show a mix of shop-welded and field-bolted assemblies.

Rate basis: Fixed rate band of \$3,500 (low) to \$4,500 (high) per fabricated ton applied. Midpoint rate of \$4,000 per fabricated ton used for the headline cost. These rates represent current USA structural steel shop fabrication pricing for the member types and complexity class identified in this package.

Tax applied: 0% for USA.

Global assumptions:

- All column lengths are assumed based on building elevation data, as they are absent from schedules.
- Weights for HSS sections not in the provided reference table have been sourced from external AISC data.

- HSS9x5x3/8 on S1.02 is assumed to be a typo for a standard size; weight for HSS10x4x3/8 was used.
- Cap plates (1/2" thick) are assumed for all HSS columns as none were specified.

3. MEMBER-BY-MEMBER TONNAGE TAKE-OFF

--- Table 3A — Primary Structural Members Columns, primary beams, moment frames, transfer members, and lateral bracing.

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (m)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
1	Beam	—	W14x22	30	12'-10"	3,912	32.7	3,838	8,462	A992	S1.02	Plan Count
2	Beam	—	W14x22	6	25'-2 1/2"	7,684	32.7	1,507	3,323	A992	S1.02	Plan Count
3	Beam	—	W16x36	1	16'-11"	5,156	53.6	276	608	A992	S1.02	Plan Count
4	Beam	—	HSS18x6x3/8	2	84'-4 7/8"	25,727	NOT INSTABLE	3,628	8,000	A500	S1.02	Plan Count
5	Beam	—	HSS18x6x3/8	2	85'-7 5/8"	26,102	NOT INSTABLE	3,680	8,114	A500	S1.02	Plan Count
6	Column	C5A	HSS6x6x5/16	8	Est. 43'-0"	13,106	NOT INSTABLE	2,999	6,612	A500	S1.01, S0.02	Plan Count

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (m)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
7	Column	C5	HSS6x6x3/8	34	Est. 43'-0"	13,106	33.1	14,756	32,532	A500	S1.01, S0.02	Plan Count
8	Column	C6	HSS6x6x5/8	4	Est. 43'-0"	13,106	NOT IN TABLE	2,176	4,797	A500	S1.01, S0.02	Plan Count
9	Column	C86	HSS8x6x3/8	2	Est. 14'-0"	4,267	NOT IN TABLE	323	712	A500	S1.02, S3.06	Plan Count
10	Column	—	HSS8x8x3/8	2	Est. 14'-0"	4,267	44.9	383	844	A500	S1.02, S3.05	Plan Count

Table 3B — Secondary and Miscellaneous Members Secondary beams, purlins, girts, joists, bracing, stairs, handrails, and platforms.

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (m)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
11	Beam	—	HSS9x5x3/8	2	16'-11"	5,156	NOT IN TABLE	416	917	A500	S1.02	Plan Count

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
12	Beam	—	HSS6x4x1/4	1	4'-2 3/4"	1,289	18.0	23	51	A500	S1.02	Plan Count
13	Beam	—	HSS8x6x3/8	1	13'-0"	3,962	NO TINTABLE	150	331	A500	S3.06	Detail
14	Beam	—	W8x15	2	25'-2 1/2"	7,684	22.3	342	754	A992	S3.05	Detail
15	Purlin	—	HSS6x4x1/4	4	11'-0"	3,353	18.0	241	531	A500	S3.05	Detail

Table 3C — Connection Material and Plates Shear plates, base plates, stiffeners, gusset plates, and embedded plates.

No.	Description	Thickness	Width	Length	Qty	Weight (kg)	Weight (lb)	Source Sheet	Extraction Method
16	C5A Base Plate	3/4"	12"	12"	8	111	245	S0.02	Schedule
17	C5 Base Plate	3/4"	12"	12"	34	472	1,041	S0.02	Schedule
18	C6 Base Plate	1"	12"	12"	4	73	161	S0.02	Schedule
19	Trellis Base Plate	1"	16"	16"	2	65	143	S3.06	Detail
20	Canopy Base Plate	1"	14"	14"	2	50	110	S3.05	Detail

No.	Description	Thickness	Width	Length	Qty	Weight (kg)	Weight (lb)	Source Sheet	Extraction Method
21	Assumed Cap Plates	1/2"	Various	Various	50	458	1,010	—	Assumed
22	Wind Post Plates	1/2"	5.5"	12"	8	32	71	S0.02	Detail

Table 3D — Tonnage Subtotals Consolidated weight by category feeding into the cost model.

Category	Count	Total Weight (kg)	Total Weight (lb)	Est. Tons
W-Shapes — Columns and Beams	39	5,963	13,147	5.96
HSS and Tube	54	28,775	63,440	28.78
Pipe	0	0	0	0.00
Angles and Channels	0	0	0	0.00
Plates (all types)	100	1,261	2,781	1.26
Miscellaneous and Embeds	0	0	0	0.00
Bolts and Accessories (3.00% addition)	—	1,080	2,381	1.08
TOTAL	193	37,079	81,749	37.08

4. COST BUILD-UP BY MEMBER CATEGORY

--- Per-category cost using the three-rate band. Adjusted rates = base rate x composite factor.

Category	Est. Tons	Adj. Rate Low (USD/ton)	Adj. Rate Mid (USD/ton)	Adj. Rate High (USD/ton)	Cost Low (USD)	Cost Mid (USD)	Cost High (USD)	Pct of Total
W-Shapes	5.96	\$3,850	\$4,400	\$4,950	\$22,900	\$26,200	\$29,500	16.1%
HSS and Tube	28.78	\$3,850	\$4,400	\$4,950	\$110,800	\$126,600	\$142,500	77.6%
Pipe	0.00	\$3,850	\$4,400	\$4,950	\$0	\$0	\$0	0.0%
Angles and Channels	0.00	\$3,850	\$4,400	\$4,950	\$0	\$0	\$0	0.0%
Plates	1.26	\$3,850	\$4,400	\$4,950	\$4,900	\$5,500	\$6,200	3.4%
Miscellaneous and Embeds	0.00	\$3,850	\$4,400	\$4,950	\$0	\$0	\$0	0.0%
Bolts and Accessories	1.08	\$3,850	\$4,400	\$4,950	\$4,200	\$4,800	\$5,300	2.9%
TOTAL	37.08	\$3,850	\$4,400	\$4,950	\$142,800	\$163,100	\$183,500	100%

5. PROCESS COST BREAKDOWN — MIDPOINT SCENARIO

--- Distributing \$163,200 across industry-standard fabrication process activities. Percentages are fixed fabrication industry standards applied at midpoint.

Fabrication cost allocated across process activities at the midpoint scenario.

Process Activity	Industry Pct	Amount (USD)	Notes
Material — mill sections and plate	35%	\$57,100	Structural steel mill cost
Shop labour — cut, fit, and weld	30%	\$49,000	Includes AWS D1.1 weld operations

Process Activity	Industry Pct	Amount (USD)	Notes
Coatings and surface preparation	12%	\$19,600	SSPC class as specified
Consumables — weld wire, bolts, and gas	6%	\$9,800	Per-ton allowance
Equipment — plasma, burn table, drill	7%	\$11,400	CNC and NC machine allocation
Overhead and QA/QC	6%	\$9,800	Inspection, NDT, project management
Shop margin	4%	\$6,500	Fabricator margin
TOTAL	100%	\$163,200	Midpoint scenario

6. DRAWING COMPLETENESS AND COST EXPOSURE ASSESSMENT

--- Areas of drawing incompleteness that create commercial exposure for this estimate.

Area of Assessment	Status	Commercial Exposure if Unresolved
All quantities confirmed from BOM or member schedule	NOT CONFIRMED	CRITICAL: Quantities derived from plan counts; schedules incomplete.
Material grades explicit on all members	CONFIRMED	PASS: Grades are listed in General Notes on S0.01.
Surface treatment confirmed, not assumed	CONFIRMED	PASS: Primary coating system is defined on S0.01.
Assembly complexity confirmed from drawings	PARTIAL	WARNING: Connection details are schematic, requiring assumption.
No conflicting BOM vs general notes grade callouts	CONFIRMED	PASS: No conflicts found.

Area of Assessment	Status	Commercial Exposure if Unresolved
Drawings are IFC or internally coordinated	NOT CONFIRMED	CRITICAL: Missing architectural drawings for lintel scope.
Galvanizing, AESS, or fire rating fully resolved	PARTIAL	WARNING: Galvanizing requirements are noted but not fully quantified.

Cost Waterfall — Midpoint Scenario Step-by-step cost build-up at the midpoint rate.

Step	Amount (USD)
Base cost — 37.08 t x \$4,400/ton	\$163,152
Tax at 0%	+\$0
Grand Total — Low (at \$3,500/ton base x composite x tax)	\$142,800
Grand Total — Mid — Headline Cost	\$163,200
Grand Total — High (at \$4,500/ton base x composite x tax)	\$183,500

7. COST CONVERSION SUMMARY

--- Final locked cost summary for this estimate.

Field	Value
Fabricated Tonnage	37.08 t
Fixed Rate Band Applied	\$3,500 — \$4,500 per ton
Composite Adjustment Factor	x1.10
Adjusted Rate — Low	\$3,850 per ton

Field	Value
Adjusted Rate — Mid	\$4,400 per ton
Adjusted Rate — High	\$4,950 per ton
Country and Tax	USA at 0%
Grand Total — Low (inc. tax)	\$142,800
Grand Total — Mid — Headline (inc. tax)	\$163,200
Grand Total — High (inc. tax)	\$183,500
Cost per Fabricated Ton — Mid	\$4,400 per ton

8. ASSUMPTIONS AND EXCLUSIONS

--- Key Assumptions (minimum five specific items — cite sheet references):

- Column lengths are not defined on schedule S0.02. A full building height of 43'-0" (13,106 mm) was assumed for all columns on grids B and C and wind posts, derived from roof elevation on S2.03.
- Rate band of \$3,500 to \$4,500 per fabricated ton represents current USA market pricing for the member types, surface treatment, and assembly complexity identified in this estimate.
- Quantities for all members were derived by counting elements on plans S1.01 and S1.02, as schedules on S0.02 lack quantities.
- HSS9x5x3/8 shown on S1.02 is assumed to be a non-standard designation; the weight for HSS10x4x3/8 was used for costing. This requires clarification.
- All HSS columns are assumed to receive a 1/2" thick cap plate for beam connections, as none are shown or specified on S0.02.

Exclusions — Items NOT included in this estimate:

- Site erection labour, cranes, manlifts, rigging, and flagging.
- Professional engineer stamping or delegated connection design.
- Loose anchor bolts cast-in-place by the general contractor.
- Site touch-up painting after final erection.
- Tax or duties beyond 0% applied in this estimate.
- 3D MEP coordination and cross-trade clash detection.

- Inflation beyond 30 calendar days from the date of this estimate.
- CRITICAL EXCLUSION: All steel lintels (W8x24, W16x50, L7x4, L6x4, L4x4) specified on S0.02 schedules are EXCLUDED from this estimate due to missing quantities and lengths, which are deferred to un-provided architectural drawings.

Change Order Triggers:

- Tonnage increases or decreases exceeding 5% from this estimate.
- Coating system upgrade — for example, from primer-only to hot-dip galvanised.
- Clarification of lintel schedule quantities and lengths from architectural drawings.
- Addition of member categories not shown on the current drawing set.
- Revision of drawings that changes profile sizes, lengths, or quantities.

10. FINAL RECOMMENDATION

--- This estimate is preliminary and carries high commercial risk due to significant gaps in the provided structural drawings. The tonnage and cost are based on multiple necessary assumptions, particularly regarding column lengths and the complete exclusion of the steel lintel package. Before this estimate is used for a binding bid, the owner must provide complete schedules or architectural drawings. The top priority is to issue RFIs to clarify all missing quantities, lengths, and non-standard profiles. If the lintel package is added or column lengths change, this estimate must be re-run with the updated inputs.

ACTION SUMMARY

--- Priority Actions for This Analysis

No.	Priority	Action Required	Owner	Deadline	Blocking
1	CRITICAL	RFI to EOR: Quantify all steel lintels from schedules on S0.02. Provide lengths, quantities, and locations.	Estimator	ASAP	Bid

No.	Priority	Action Required	Owner	Deadline	Blocking
2	CRITICAL	RFI to EOR: Provide all column lengths in the Steel Column Schedule on S0.02. Confirm if columns are multi-story.	Estimator	ASAP	Bid
3	WARNING	RFI to EOR: Confirm weights and availability for non-standard HSS sections (HSS18x6x3/8, HSS9x5x3/8, etc.).	Estimator	ASAP	Material Order
4	REVIEW	RFI to EOR: Confirm connection design intent. Is it simple shear, moment, or a mix? Provide typical details.	Estimator	ASAP	Detailing